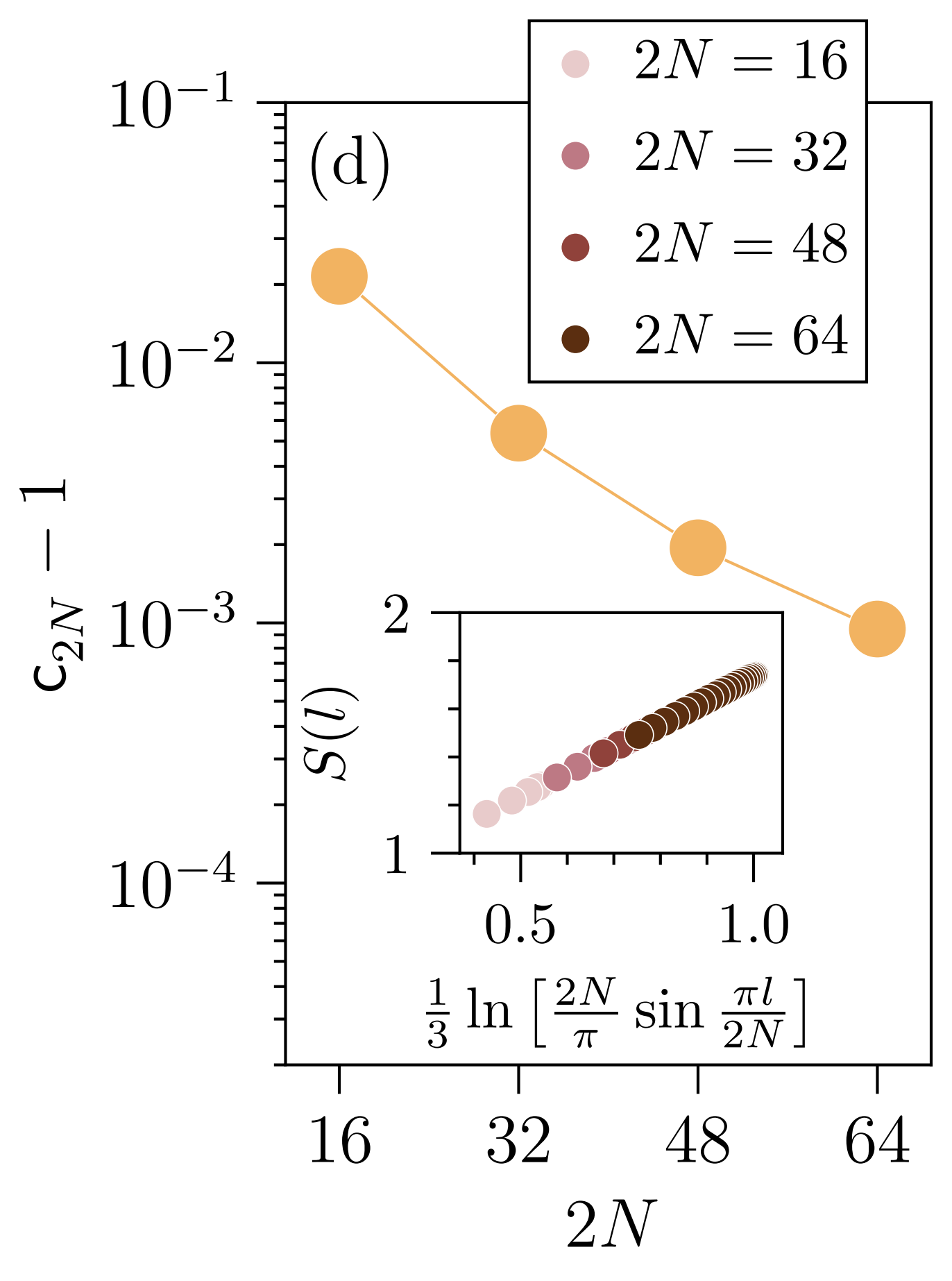

(d)
$2N = 16$
$2N = 32$
$2N = 48$
$2N = 64$
10^{-1}
10^{-2}
10^{-3}
10^{-4}
$c_{2N} - 1$
16
32
48
64
$2N$
$S(l)$
2
1
0.5
1.0
$\frac{1}{3} \ln \left[\frac{2N}{\pi} \sin \frac{\pi l}{2N} \right]$